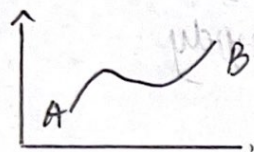


第二型曲线积分 —— 四类



“变力沿曲线做功”

$$F = \{P, Q\} \quad \begin{matrix} \uparrow Q \\ \rightarrow P \end{matrix}$$

$$\int_L P dx + Q dy \quad (+ R dz)$$

一. 一投二代三计算

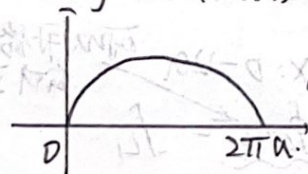
$$\begin{cases} x = x(t) \\ y = y(t) \end{cases} \quad (t, \alpha \rightarrow \beta) \text{ 只对应, 无大小.}$$

$$\int_L P(x, y) dx + Q(x, y) dy = \int_{\alpha}^{\beta} [P(x(t), y(t)) \cdot x'(t) + Q(x(t), y(t)) \cdot y'(t)] dt$$

例 18.29

$$\int_L (2a - y) dx - (a - y) dy, \text{ 其中 } L \text{ 为摆线}$$

$$\begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases} \quad \text{第一类, } t: 0 \rightarrow 2\pi$$



$$\int_0^{2\pi} (2a - a(1 - \cos t)) \cdot a(1 - \cos t)' - (a - a(1 - \cos t)) \cdot a \sin t' dt$$

$$x'(t) = a(1 - \cos t)$$

$$y'(t) = a \sin t$$

2. 格林公式

dy 前的对 x 求偏导

$$\oint_L P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) d\sigma$$

① 曲线封闭 内部无奇点



例 18.31

$$\oint_L (y - e^x) dx + (3x + e^y) dy, \quad L: x^2 + y^2 = 1 \quad \text{正向一周}$$

$$\text{原式} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy \quad r^2 = r - r \cos \theta$$

$$r = 1 - \cos \theta$$

$$S = \frac{3}{2} \pi a^2 \quad (\text{心脏线面积})$$



② 曲线封闭 内部有奇点 —— 挖洞法

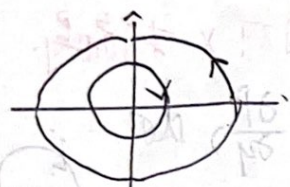
$$\oint_L P dx + Q dy = \oint_{L_1} P dx + Q dy - \oint_{L_2} P dx + Q dy$$



例 18.32

$$\text{求曲线积分} \int_L \frac{(x-y) dx + (x+y) dy}{x^2 + y^2}, \text{ 其中}$$

$$L \text{ 为沿椭圆 } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ 逆时针一周 } a, b > 0.$$



$$\int_L \frac{(x-y)dx + (x+y)dy}{x^2+y^2} = \int_{L_1} \frac{(x-y)dx + (x+y)dy}{x^2+y^2} \quad \text{where } x^2+y^2=r^2$$

$$= \frac{1}{r^2} \oint_P \frac{(x-y)dx + (x+y)dy}{r^2} = \frac{1}{r^2} \oint_P \frac{\partial Q}{\partial x} dx + \frac{\partial P}{\partial y} dy = \frac{1}{r^2} \oint_P 2 d\sigma = 2\pi$$

平面旋度为0

② 非封闭曲线 $\rightarrow$ 换路径

例 18.33

求 $\int_L \frac{(x-y)dx + (x+y)dy}{x^2+y^2}$, L 为从点 $A(\frac{\pi}{2}, 0)$ 到点 $B(\frac{\pi}{2}, 0)$ 的曲线 $y = \cos x$.



条件 ① 非封闭 $\odot \frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$

$\Rightarrow$ 换路径: 可以把路径换成圆

$$\rightarrow x^2+y^2 = \frac{\pi^2}{4}$$

$$I = \int_{L_1} \frac{(x-y)dx + (x+y)dy}{x^2+y^2} = \frac{4}{\pi} \int_{L_1} (x-y)dx + (x+y)dy$$

参数法 $\begin{cases} x = \frac{\pi}{2} \cos \theta \\ y = \frac{\pi}{2} \sin \theta \end{cases}$

$$I = \frac{4}{\pi} \int_0^{2\pi} \left[\frac{\pi}{2} (\cos \theta - \sin \theta) \cdot \frac{\pi}{2} (-\sin \theta) + \frac{\pi}{2} (\cos \theta + \sin \theta) \cdot \frac{\pi}{2} \cos \theta \right] d\theta$$

$$= \int_0^{2\pi} -\cos \theta \sin \theta + \sin \theta + \cos \theta + \sin \theta \cos \theta d\theta$$

$$= \int_0^{2\pi} d\theta = 2\pi$$

④ 非封闭曲线 $\rightarrow$ 加线减线法

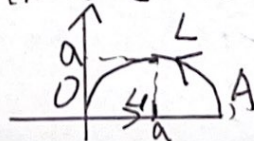
例 18.34

求 $\int_L [e^x \sin y - b(x+y)] dx + (e^x \cos y - ax) dy$.

其中 a, b 为常数, L 为沿上半圆 $y = \sqrt{2ax - x^2}$ 从点 $A(2a, 0)$ 到点 $O(0, 0)$.

$x^2+y^2-2ax=0$ 补 $L_1: y=0, x: 0 \rightarrow 2a$. 可用格林公式了.

$$(x-a)^2+y^2=a^2 \quad I = \int_L = \int_{L_1} + \int_{L_2} - \int_{L_3}$$



$$\frac{\partial P}{\partial y} = e^x \cos y - b$$

$$\frac{\partial Q}{\partial x} = e^x \cos y - a$$

$$\begin{aligned} \oint_{L^+} &= \iint_D (e^x \cos y - a) - (e^x \cos y - b) d\sigma \\ &= (b-a) \iint_D d\sigma = (b-a) \frac{\pi}{2} a^2 \\ \int_{L^+} -bx dx &= \int_0^{2a} -bx dx = -b \frac{(2a)^2}{2} \\ \text{当 } y=0 \text{ 时 } \quad S &= (b-a) \frac{\pi}{2} a^2 + b \frac{(2a)^2}{2} \end{aligned}$$

⑤ 积分与路径无关.

- $\int_{L^+} P(x,y) dx + Q(x,y) dy$ 与路径无关
- $Pdx + Qdy$ 为某二元函数 $u(x,y)$ 的全微分.
- $Pdx + Qdy = 0$ 为全微分方程.
- $Pi + Qj$ 为 $u(x,y)$ 梯度
- D 为任何分段光滑闭曲线 $\oint_L Pdx + Qdy = 0$.
- $\boxed{\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}}$ D 内处处成立

求积分 { ① 折线法
② 凑微分法

① 折线法 $u(x,y) = \int_{(x_0,y_0)}^{(x,y)} P(x,y) dx + Q(x,y) dy + u(x_0,y_0)$

$$u(x,y) = \int_{x_0}^x P(x, y_0) dx + \int_{y_0}^y Q(x, y) dy + u(x_0, y_0)$$

② 凑微分. (技巧性强)

例 18.36

验证 $\int_{(10,0)}^{(\frac{\pi}{2}, 1)} (2xy^3 - y^2 \cos x) dx + (1 - 2y \sin x + 3x^2 y) dy$

$$\begin{aligned} \text{原式} &= \int_0^{\frac{\pi}{2}} (2xy^3 - y^2 \cos x) dx \Big|_{y=0} + \int_0^1 (1 - 2y \sin x + 3x^2 y) dy \Big|_{x=\frac{\pi}{2}} \\ &= \frac{\pi^2}{4} \end{aligned}$$

先验证 $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

$$\begin{aligned} \text{证: } &(2xy^3 - y^2 \cos x) dx + (1 - 2y \sin x + 3x^2 y) dy \\ &= 2xy^3 dx + 3x^2 y dy - (y^2 \cos x dx + 2y \sin x dy) + dy \\ &= d(x^2 y^3) - d(y^2 \sin x) + dy \\ &= d(x^2 y^3 - y^2 \sin x + y) \\ \text{故原式} &= (x^2 y^3 - y^2 \sin x + y) \Big|_{(10,0)}^{(\frac{\pi}{2}, 1)} = \frac{\pi^2}{4} \end{aligned}$$

例 18.17

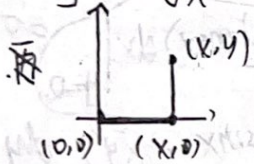
设 $f(x)$ 有连续二阶导数, $f(0)=0$, $f'(0)=1$.

且 $\int_L [xy(x+y) - f(x)] dx + [f(x) + x^2y] dy$
与路径无关, 求 $f(x)$.

$$\int_{(0,0)}^{(x,y)} [xy(x+y) - f(x)] dx + [f(x) + x^2y] dy$$

[solution]

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow \begin{cases} f(x) = \frac{1}{12}x^4 + C_1x + C_2 \\ f(0)=0 \\ f'(0)=1 \end{cases}$$



$$\begin{aligned} & \int_{(0,0)}^{(x,y)} [xy(x+y) - f(x)] dx + [f(x) + x^2y] dy \\ &= - \int_0^x (\frac{x^4}{12} + x) dx + \int_0^y (1(\frac{x^3}{3} + 1) + x^2) dy \end{aligned}$$

例 18.38 ★

$f(0)=0$, $g(0)=0$. 对于平面上任意闭曲线 L .

$$\oint_L [yf(x) + 2ye^x + 2yg(x)] dx + 2[yg(x) + f(x)] dy = 0.$$

(1) 求 $f(x)$, $g(x)$ 表达式.

(2) 设 L_1 任一条从 $(0,0)$ 到 $(1,1)$ 曲线.

利用 $f(x)$, $g(x)$ 计算.

$$\int_{L_1} [yf(x) + 2ye^x + 2yg(x)] dx + 2[yg(x) + f(x)] dy$$

[solution]

$$P(x,y) = y^2f(x) + 2ye^x + 2yg(x)$$

$$Q(x,y) = 2yg(x) + 2f(x)$$

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow \begin{aligned} 2yf(x) + 2e^x + 2g(x) &= 2yg'(x) + 2f'(x) \\ \Rightarrow y[g'(x) - f(x)] &= -[f'(x) - g(x) - e^x] \end{aligned}$$

比较 y 的同幂项系数.

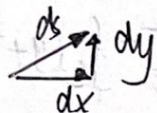
$$\begin{cases} g'(x) - f(x) = 0 \\ f'(x) - g(x) - e^x = 0 \end{cases} \Rightarrow \begin{aligned} g''(x) - g(x) &= e^x \\ g(x) &= C_1e^x + C_2e^{-x} + \frac{1}{2}xe^x \end{aligned}$$

13. 两类曲线积分的关系

$$\int_L P dx + Q dy = \int_L (P \cos \alpha + Q \sin \alpha) ds.$$

其中 $(\cos \alpha, \sin \alpha)$ 为 L 上点 (x, y) 处与 L 同向的

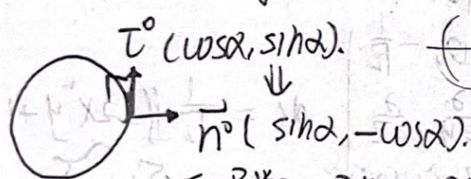
单位切向量.



例 18.42 好久没考了.

设 L 为曲线 $x^2 + y^2 = R^2$ (常数 $R > 0$) 按逆时针绕一周, n 为 L 的外法线方向向量.

$$u(x, y) = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = x + y, \text{ 求 } \oint_L \frac{\partial u}{\partial n} ds$$



$$\therefore \text{方向导数 } \frac{\partial u}{\partial n} = \frac{\partial u}{\partial x} \sin \alpha + \frac{\partial u}{\partial y} (-\cos \alpha)$$

$$\oint \frac{\partial u}{\partial n} ds = \oint_L \left(\frac{\partial u}{\partial x} \sin \alpha - \frac{\partial u}{\partial y} \cos \alpha \right) ds$$

$$= \oint_L \frac{\partial u}{\partial x} dy - \frac{\partial u}{\partial y} dx \xrightarrow{\text{格林}} \iint_D \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) d\sigma$$

$$= \iint_D (x + y) d\sigma = \int_0^{2\pi} d\theta \int_0^R r^3 dr = \frac{\pi}{2} R^4$$

$$D = \{(x, y) | x^2 + y^2 \leq R^2\}$$

14. 空间问题.

$$\textcircled{1} \int_L P dx + Q dy + R dz = \int_a^b P[x(t), y(t), z(t)] x'(t) dt + Q[x(t), y(t), z(t)] y'(t) dt + R[x(t), y(t), z(t)] z'(t) dt$$

$$\begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases}$$

例 18.43

$$\text{计算 } I = \oint_{\Gamma} (y^2 + z^2) dx + (z^2 + x^2) dy + (x^2 + y^2) dz.$$

Γ 是球面 $x^2 + y^2 + z^2 = 2RX$ 与柱面 $x^2 + y^2 = 2ax$ 的交线. ($0 < a < R, 2 > 0$).

而 Γ 按此方向进行, 使它在球的外表面上所围的区域 S 在其上方.

$$\begin{aligned} (x+y)^2 + z^2 &= 2RX \\ x^2 + y^2 &= 2ax \rightarrow r = 2a \cos \theta \\ z &= \sqrt{2(R-a)x} \end{aligned}$$

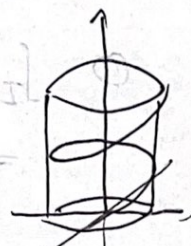
$$\text{用参数方程: } \begin{cases} x = r \cos \theta = 2a \cos^2 \theta \\ y = r \sin \theta = 2a \cos \theta \sin \theta \\ z = \sqrt{2(R-a)r \cos \theta} = \sqrt{2(R-a)2a \cos^3 \theta} \end{cases}$$

$$I = \int_{-\pi/2}^{\pi/2} \bigcirc d\theta$$

例 18.44

计算 $\int_C y dx + z dy + x dz$, Γ 为螺旋线

$$\begin{cases} x = a \cos t \\ y = a \sin t \\ z = bt \end{cases} \quad \text{从 } t=0 \text{ 到 } t=2\pi \text{ 的一段}$$



$$\int_0^{2\pi} a \sin t \cdot a(1 - \sin t) + b t \cdot a \cos t + a \cos t \cdot b \, dt$$

例 18.4

曲线闭且闭, 在同一平面上或投影简单.

用斯托克斯公式.

$$\oint_C P dx + Q dy + R dz = \iint_{\Sigma} \begin{vmatrix} dy dz & dz dx & dx dy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} \quad (\text{第一型曲面积分})$$

$$\iint_{\Sigma} \begin{vmatrix} \cos \alpha & \cos \beta & \cos \gamma \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} dS \quad (\text{第一型曲面积分})$$

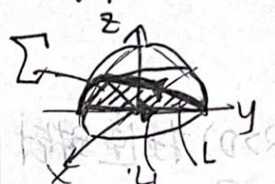
若为平面
最方便

例 18.4b.

已知曲线 $L: \begin{cases} z = \sqrt{2-x^2-y^2} \\ z = x \end{cases}$, 起点为 $A(0, \sqrt{2}, 0)$.

终点为 $B(0, -\sqrt{2}, 0)$

计算曲线积分 $I = \int_L (y+z) dx + (z^2-x^2+y) dy + x y^2 dz$



$$= \oint_{L+L_1} - \int_{L_1}$$

$$\textcircled{1} \quad F = x - z = 0 \quad \vec{n} = (1, 0, -1) \quad \vec{n}_0 = \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}} \right)$$

$$\iint_{\Sigma} \begin{vmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y+z & z^2-x^2+y & x y^2 \end{vmatrix} dS = \frac{1}{\sqrt{2}} \iint_{\Sigma} (2xy+1) dS \rightarrow 0$$

$$\oint_{L+L_1} (y+z) dx + 0 dy + 0 dz = \frac{1}{\sqrt{2}} \iint_{\Sigma} dS$$

$$\textcircled{2} \quad \int_{L_1} (y+z) dx + 0 dy + 0 dz = \int_{-\sqrt{2}}^{\sqrt{2}} y dy = 0$$

$$I = \frac{\sqrt{2}}{2} \pi$$

③ $\text{rot } F=0$ (无旋场), 可换路径

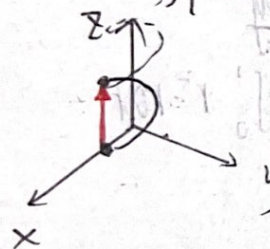
例 ★ 18.47 (2022).

设 Γ 是圆柱螺旋线, $X=R\cos\theta$

$$\begin{cases} Y=R\sin\theta \\ Z=b\theta \end{cases} \quad R>0.$$

从 $A(R,0,0)$ 到 $B(R,0,2\pi b)$.

求 $J = \int_{\Gamma} (x^2 - yz)dx + (y^2 - zx)dy + (z^2 - xy)dz$.



$$\text{rot } \vec{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 - yz & y^2 - zx & z^2 - xy \end{vmatrix} = \vec{0}.$$

$X=R \quad Y=0 \quad Z: 0 \rightarrow 2\pi b.$

$$J = \int_{\Gamma} z^2 dz = \int_0^{2\pi b} z^2 dz.$$

五. 第二型曲面积分.

1) 通量

2) 计算

① 一投二代三计算

分别投到 yOz xOy xOz 三个平面

习惯

a. 投影为线 $\rightarrow$ 积分为 0.

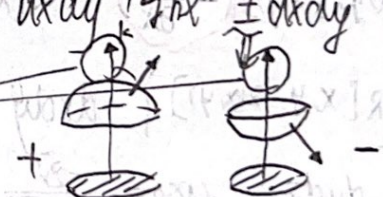
b.  分成上下曲面分别计算

c. 将 Σ 写成 $z(x,y)$. 计算

(一投) 二代: $z \rightarrow z(x,y)$

三计算: 把 $dxdy$ 写成 $\pm dxdy$

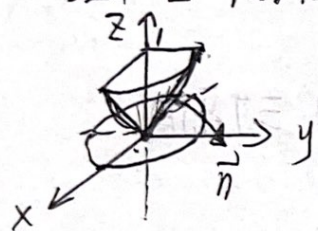
题目告知:



例 18.49 ★

计算. $I = \iint_{\Sigma} x dy dz + y dz dx + z dx dy$

其中 Σ 取曲面 $z = x^2 + y^2$ 在第一象限, $0 \leq z \leq 1$



$$\iint_{\Sigma} x dy dz = \iint_{D_1} \sqrt{z-y^2} dy dz = \frac{\pi}{8}$$

$$\iint_{\Sigma} y dz dx = \frac{\pi}{8}$$

$$\iint_{\Sigma} z dx dy = \iint_{D_2} (x^2 + y^2) dx dy = \frac{\pi}{8}$$

转换投影法

$$\iint_{\Sigma} P(x,y,z) dy dz + Q(x,y,z) dz dx + R(x,y,z) dx dy$$

$$= \pm \iint_{D_{xy}} \left\{ P(x,y,z(x,y)) \left(-\frac{\partial z}{\partial x} \right) + Q(x,y,z(x,y)) \left(-\frac{\partial z}{\partial y} \right) + R(x,y,z(x,y)) \right\} dx dy$$

$$\begin{pmatrix} ds \cos \alpha = dy dz \\ ds \cos \beta = dz dx \\ ds \cos \gamma = dx dy \end{pmatrix} \Rightarrow \begin{pmatrix} \cos \alpha = \frac{-z'_x}{\sqrt{1+z_x'^2+z_y'^2}} \\ \cos \beta = \frac{-z'_y}{\sqrt{1+z_x'^2+z_y'^2}} \\ \cos \gamma = \frac{1}{\sqrt{1+z_x'^2+z_y'^2}} \end{pmatrix} \Rightarrow \frac{dy dz}{-z'_x} = \frac{dz dx}{-z'_y} = \frac{dx dy}{1}$$

左题采用转换投影法

$$I = \iint_{\Sigma} x dy dz + y dz dx + z dx dy$$

$$= \iint_{D_1} [x \cdot (-2x) + y \cdot (-2y) + z] dx dy$$

$$= \iint_{D_1} [-2x^2 - 2y^2 + z] dx dy$$

$$= \iint_{D_{xy}} (-2x^2 - 2y^2 + x^2 + y^2) dx dy$$

$$= \iint_{D_{xy}} (-x^2 - y^2) dx dy = -\int_0^{\frac{\pi}{2}} d\theta \int_0^1 r^2 \cdot r dr = -\frac{\pi}{8}$$

例

若 $f(x,y,z)$ 连续

$$I = \iint_{\Sigma} (yf + x) dy dz + (xf + y) dz dx + (zxf + z) dx dy$$

$$z = \frac{1}{2}(x^2 + y^2) \quad z=2 \quad z=8$$

$$\Rightarrow \frac{\partial z}{\partial x} = x \quad \frac{\partial z}{\partial y} = y$$

$$I = \iint_{\Sigma} [-(yf+x)x + -(xf+y)y + zxf + z] dx dy$$

$$= \iint_{\Sigma} (z - x^2 - y^2) dx dy = \frac{1}{2} \iint_{\Sigma} z dx dy$$

13. 高斯公式

① 封闭, 内部无奇点, 直接高斯

$$\oint_{\Sigma} P dy dz + Q dz dx + R dx dy = \iiint_{\Omega} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dV$$

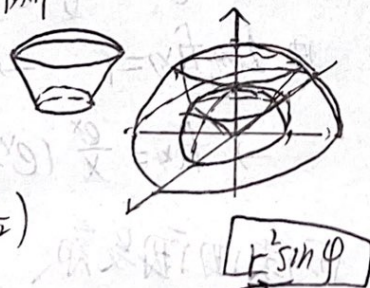
例 18.55

$$I = \oint_{\Sigma} \left[\frac{1}{y+3} f\left(\frac{x+4}{y+3}\right) + 3xy^2 \right] dy dz + \left[\frac{1}{x+4} f\left(\frac{x+4}{y+3}\right) + 3x^2y \right] dz dx + z^3 dx dy$$

其中 $f(u)$ 具有连续导数, Σ 为锥面 $z = \sqrt{x^2 + y^2}$

与两半球面 $z = \sqrt{1 - x^2 - y^2}$, $z = \sqrt{4 - x^2 - y^2}$

所围立体的全表面外侧



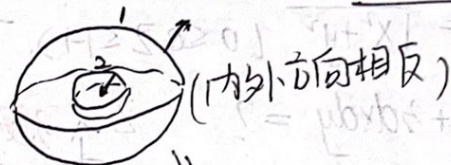
$$\frac{\partial P}{\partial x} = f' \cdot \frac{1}{y+3} + 3y^2$$

$$\frac{\partial Q}{\partial y} = 3x^2 + f' \left(-\frac{x+4}{(y+3)^2} \right)$$

$$\frac{\partial R}{\partial z} = 3z^2$$

$$I = \iiint_{\Sigma \cup \Sigma_1} (x^2 + y^2 + z^2) dV = 3 \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} \sin \varphi d\varphi \int_1^2 r^4 dr = \frac{9\pi}{5} (2^5 - 1)$$

② 封闭, 有奇点, $\text{div} F = 0$ 换个面



把 1 换成 2

例 18.56

计算 $\oint_{\Sigma} \frac{x dy dz + y dz dx + z dx dy}{(x^2 + y^2 + z^2)^{3/2}}$, 其中

Σ 为椭球面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, 法向量指向外.

$$\text{可得 } \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 0. \quad \text{一无源场}$$

$$\Sigma_1: x^2 + y^2 + z^2 = r^2$$

$$\Rightarrow \oint_{\Sigma} = \oint_{\Sigma_1} = \frac{1}{r^3} \oint_{\Sigma_1} x dy dz + y dz dx + z dx dy = \frac{1}{r^3} \iiint_{\Sigma_1} \text{div} F dV$$

③ 非封闭曲面, $\text{div} F = 0$, 换个面

例 18.57. ★

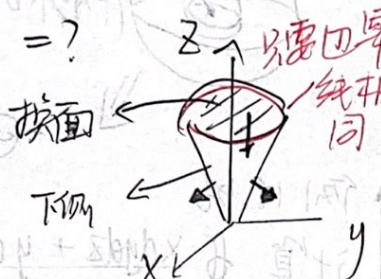
设 Σ 为锥面 $z = \sqrt{x^2 + y^2}$ ($0 \leq z \leq 1$). 下侧.

则 $\iint_{\Sigma} dydz + 2dzdx + 3dxdy = ?$

$$\left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 0 \right)$$

$$\Rightarrow \Sigma_2: z=1, x^2+y^2 \leq 1$$

$$\Rightarrow \iint_{\Sigma} dxdy = - \iint_{D_{xy}} dxdy = -3\pi$$



④ 非封闭, 补面使其封闭.

例 18.58.

$u = z(x^2 + y^2)$. 向量场 $A = \text{grad } u$ 穿过球面

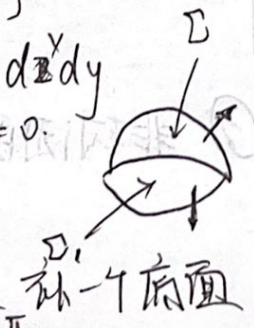
$x^2 + y^2 + z^2 = 1$ 上半球面 求向外侧通量 $Q =$

[为求] $\vec{A} = \text{grad } u = \{ 2xz, 2yz, x^2 + y^2 \}$

$$Q = \iint_{\Sigma} 2xz dydz + 0 + x^2 dzdy$$

$$Q = \iint_{\Sigma_{\text{上}} + \Sigma_{\text{下}}} - \iint_{\Sigma_{\text{下}}} x^2 dzdy$$

$$= \iiint_{\Omega} (2z) dv - \iint_{\Sigma_{\text{下}}} x^2 dzdy = \frac{1}{4}\pi$$



⑤ 由 $\text{div } F = 0$. 建方程求 $f(x)$

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 0 \rightarrow \text{得到 } f(x) \text{ 的微分方程.}$$

例 18.59

设对于 $x > 0$ 半空间内任意光滑有向曲面 Σ .

$$\text{都有 } \oint_{\Sigma} x f(x) dydz - xy f(x) dzdx - e^{2x} z dxdy = 0.$$

其中 $f(x)$ 在 $(0, +\infty)$ 为具一阶连续导数, $\lim_{x \rightarrow +\infty} f(x) = 1$.

$$\text{分析可知 } \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 0.$$

$$\Rightarrow f(x) + (\frac{1}{x} - 1)f(x) = \frac{1}{x} e^{2x}, x > 0.$$

$$f(x) = e^{-\int \frac{1}{x} dx} \left[\int \frac{1}{x} e^{2x} dx + C \right] = \frac{e^x}{x} (e^x + C)$$

$$\lim_{x \rightarrow +\infty} f(x) = 1 \rightarrow C = -1$$

$$\Rightarrow f(x) = \frac{e^x}{x} (e^x - 1), (x > 0)$$

4. 两类曲面积分

$$\begin{aligned} & \iint_{\Sigma} P dydz + Q dzdx + R dxdy \\ &= \iint_{\Sigma} (P \cos \alpha + Q \cos \beta + R \cos \gamma) ds \end{aligned}$$